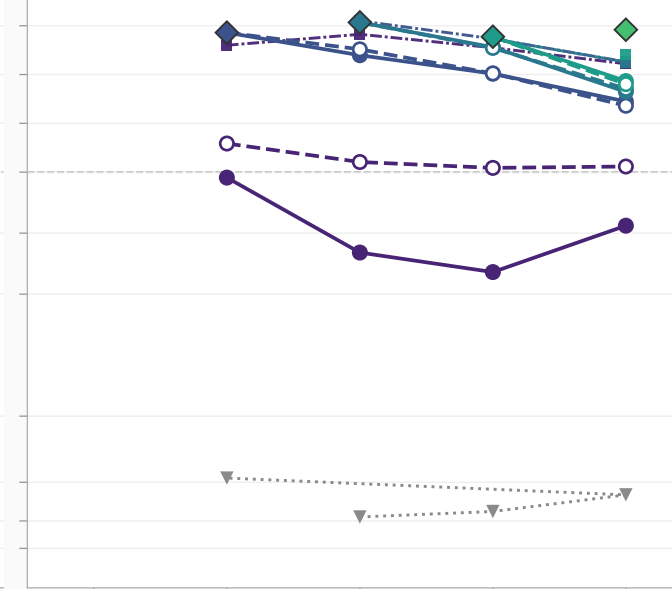
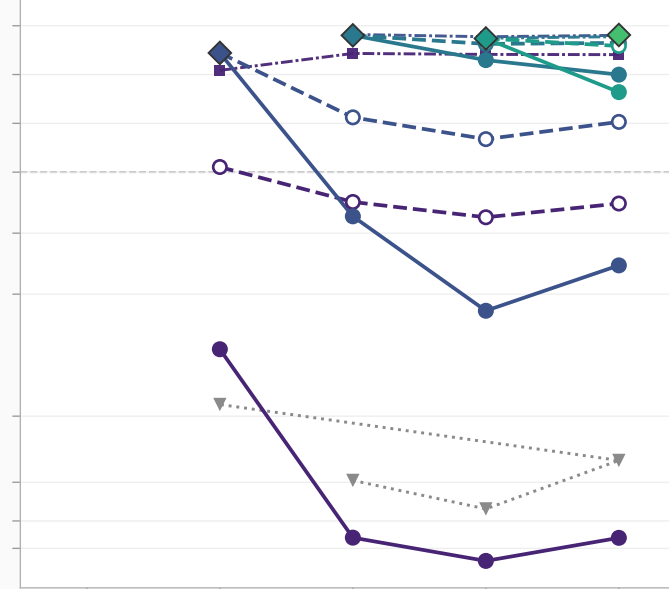
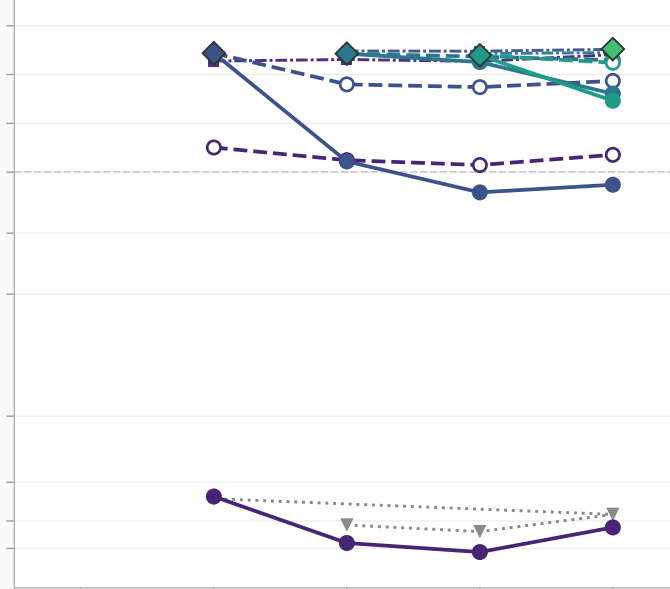
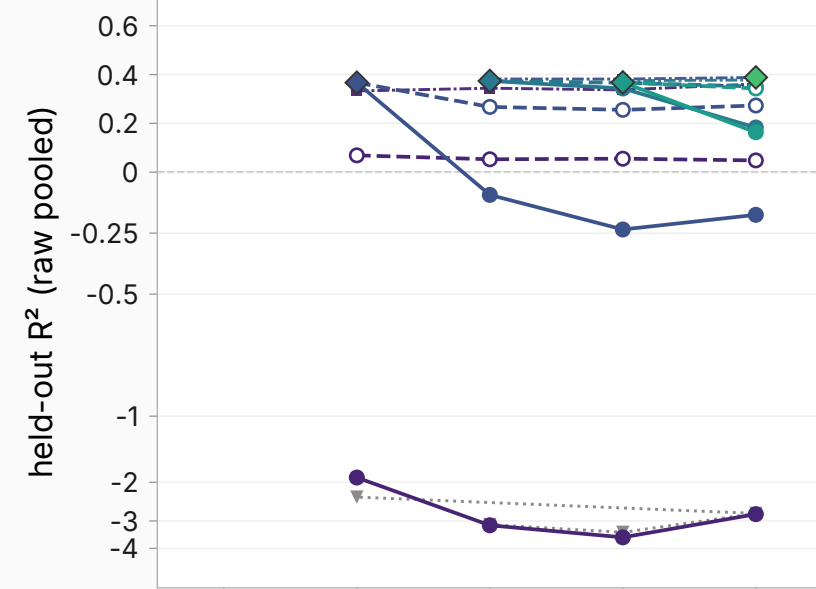


GSM8K test (n<d companion)

GSM8K train

MATH

IF-constraints

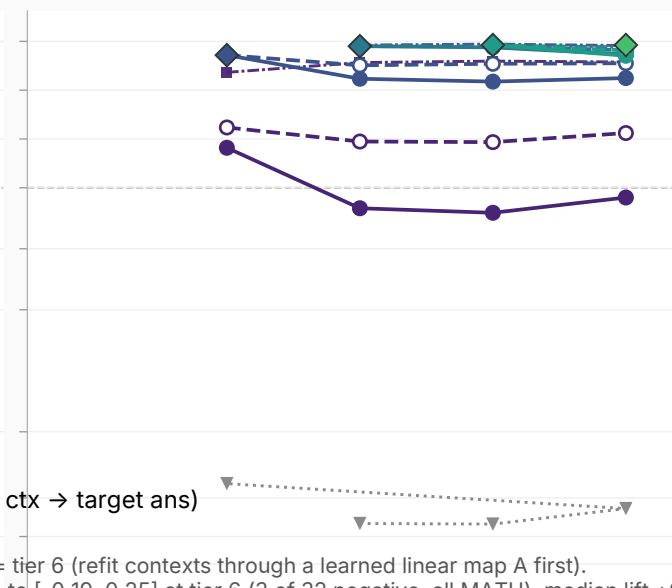
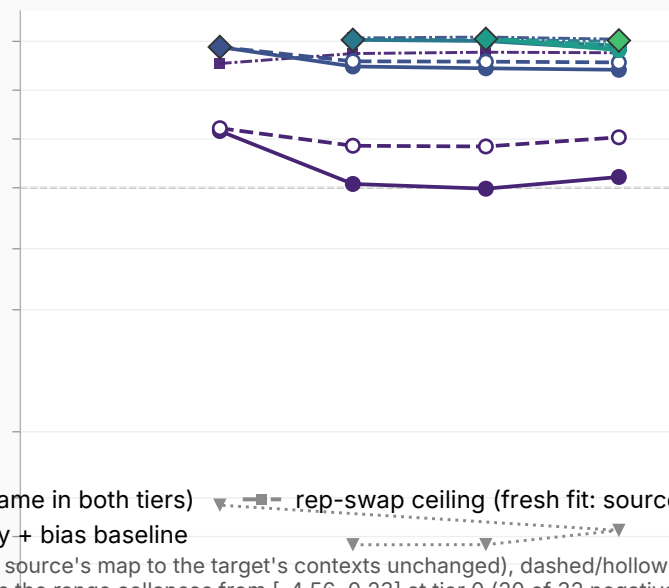
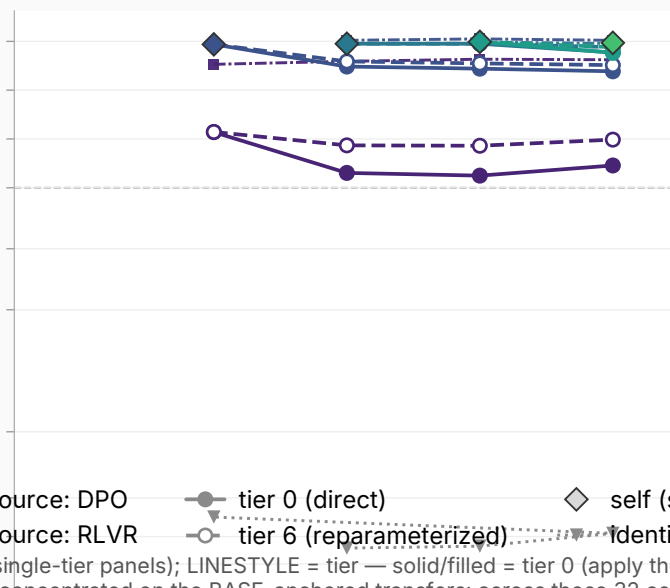
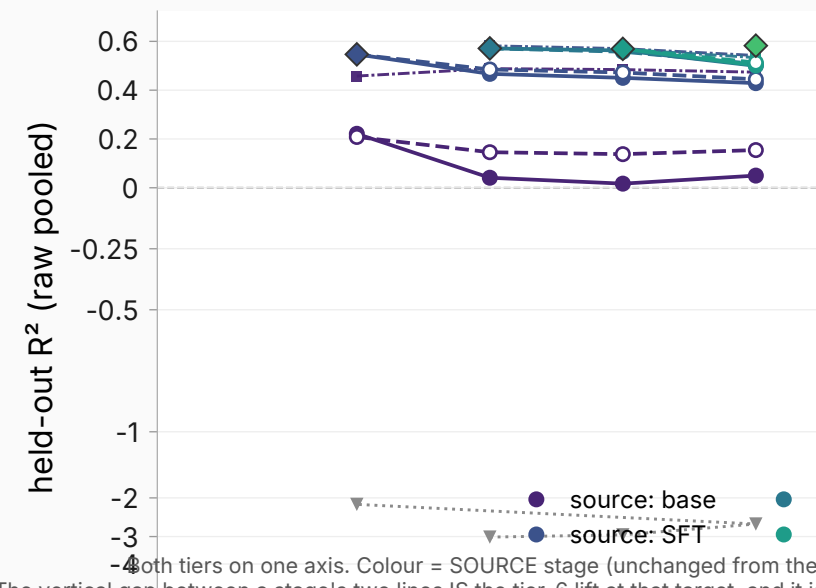


UltraFeedback

Tulu-SFT mix

LMSYS chat

LMSYS natural.



source: base

source: SFT

source: DPO

source: RLVR

tier 0 (direct)

tier 6 (reparameterized)

self (same in both tiers)

identity + bias baseline

rep-swap ceiling (fresh fit: source ctx → target ans)

base

SFT

DPO

RLVR

longer RLVR

base

SFT

DPO

RLVR

longer RLVR

base

SFT

DPO

RLVR

longer RLVR

base

SFT

DPO

RLVR

longer RLVR

longer RLVR

Both tiers on one axis. Colour = SOURCE stage (unchanged from the single-tier panels); LINESTYLE = tier — solid/filled = tier 0 (apply the source's map to the target's contexts unchanged), dashed/hollow = tier 6 (refit contexts through a learned linear map A first). The vertical gap between a stage's two lines IS the tier-6 lift at that target, and it is concentrated on the BASE-anchored transfers: across those 32 cells the range collapses from [-4.56, 0.23] at tier 0 (20 of 32 negative) to [-0.19, 0.25] at tier 6 (3 of 32 negative, all MATH), median lift +0.27. The 24 adjacent-stage cells move far less (median +0.02), but not uniformly: 8 of 24 shift by more than 0.05, the largest being SFT→DPO on the math corpora (+0.41 MATH, +0.36 GSM8K test, +0.32 GSM8K train) — visible as the one navy pair that separates. Layer 31 Llama-3.1-8B Tulu ladder is symlog below -1 (worst-plotted value -4.56). All pairs are on-policy, so even point mixes representation change with answer-distribution changes. The diamond at stage T is the source=T, target=T cell and is IDENTICAL in both tiers (a self map needs no reparameterization); grey triangles = identity+bias, also tier-invariant. 10 of the 20 ordered stage pairs now exist: SFT→longer RLVR, SFT→longer RLVR and RLVR→longer RLVR were never run for the original figure and are measured here, so only the base column is still missing data; the tier-6 lift statistics above cover the ORIGINAL pairs.